

ELECTROMAGNETIC INDUCTION

Important formulas and concepts directly from the NCERT-

❖ Magnetic flux-

- Magnetic flux through a plane of area A placed in uniform magnetic field B ,
$$\phi_B = B.A = BA\cos\theta$$
- SI unit – weber(Wb) or tesla meter squared(Tm^2)
- It is a 'scalar quantity'.

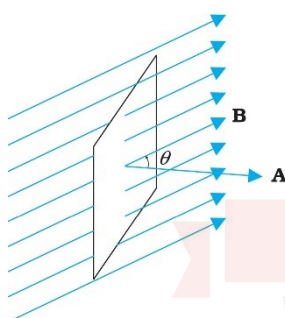


FIGURE 6.4 A plane of surface area A placed in a uniform magnetic field B .

❖ FARADAY's law of induction-

- “The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit”.
- Mathematically, induced emf is given by,
$$\varepsilon = -d\phi / dt$$

neg. sign indicates the direction of induced emf and hence current in a closed loop.
- In case of N turns,
$$\varepsilon = -Nd\phi / dt$$

the induced emf can be increased by increasing the no. of turns N of a closed coil.

❖ Lenz's Law-

- “the polarity of induced emf is such that it tends to produce a current which opposes change in flux that produced it”.

- It is based on **'law of conservation of energy'**.

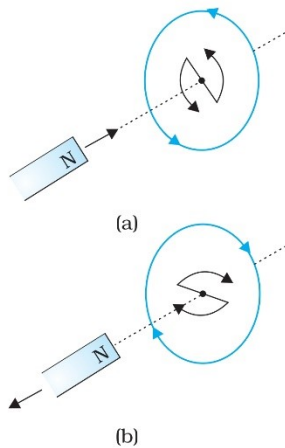


FIGURE 6.6
Illustration of
Lenz's law.

❖ **Motional Electromotive Force-**

- We can also induce emf by moving a conductor instead of varying the mag. Field that is by changing the magnetic flux .

The induced emf in this case is known as 'motional emf'

$$\varepsilon = Bvl$$

- Time varying magnetic field generates electric field.
- If r be the resistance of movable arc, then current I in the loop is,

$$I = Bvl / r$$

- Due to magnetic field, there will be a force on the movable arm which is given by,

$$F = IlB = B^2 l^2 v / r$$

- Power,

$$P = Fv = B^2 l^2 v^2 / r$$

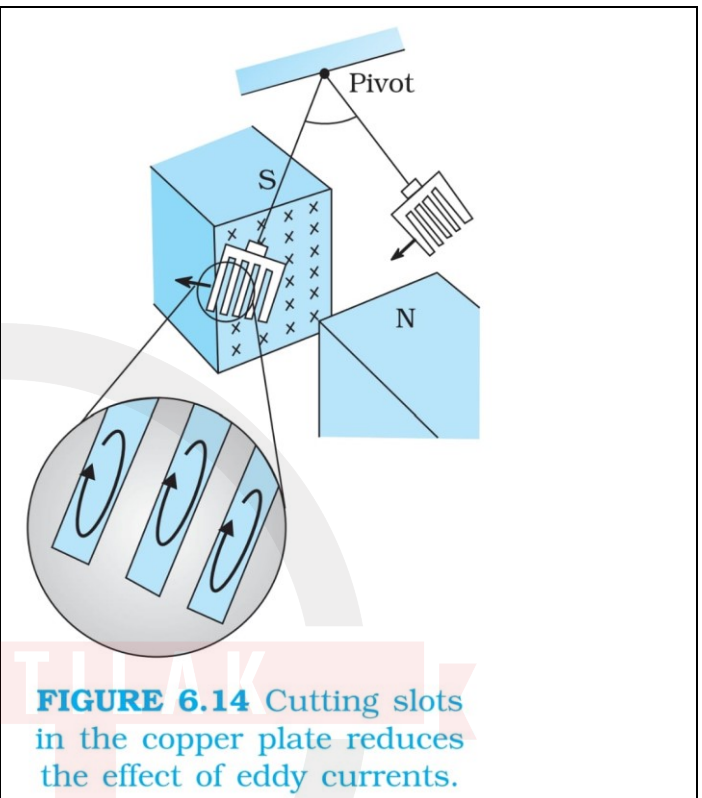
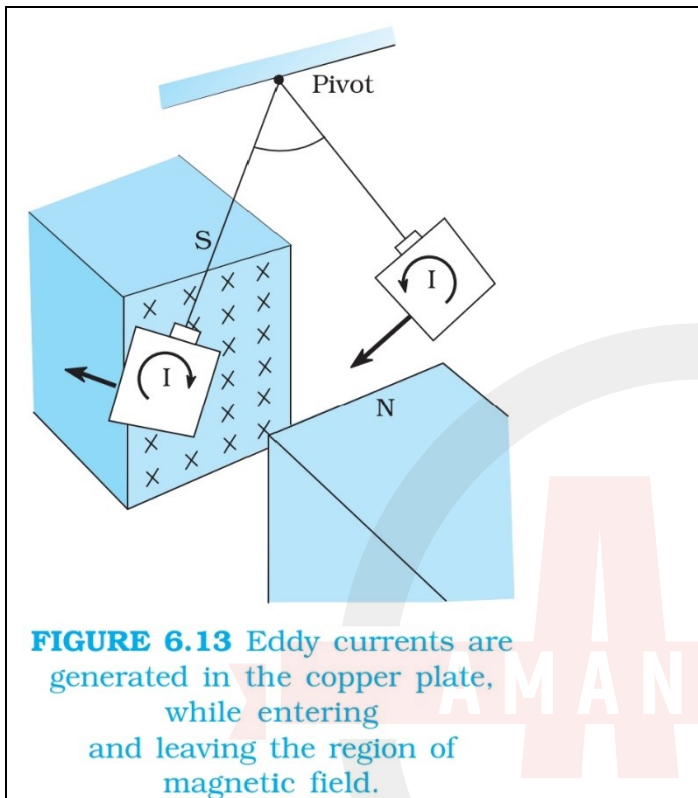
- Relation b/w charge flow through the circuit and change in magnetic flux,

$$\Delta Q = \Delta \phi_B / r$$

❖ **Eddy currents-**

- When **bulk pieces** of conductors are subjected to changing mag. Flux, induced currents produced in them . Their flow pattern resemble **swirling eddies in water**, these currents are known as 'Eddy Currents'.

- If rectangular slots are made in copper plates, area available to the flow of eddy currents is less, thus effect of eddy currents is reduced.



- Eddy currents are undesirable since they heat up the core and dissipate electrical energy in form of heat.
- Eddy currents are minimised by using laminations of metal to make a metal core. The laminations are separated by an insulating material like 'Lacquer'. The plane of the laminations must be arranged parallel to the mag. Field, so reduces the strength of eddy currents.

➤ Applications-

- 1) Magnetic braking in trains
- 2) Electromagnetic damping
- 3) Induction furnace
- 4) Electric power meters

❖ Inductance-

- The flux through a coil is proportional to current ,

$$\phi_B \propto I$$
- For a coil of **N** turns, the same flux is linked with all the turns.
 A term '**flux linkage**' is used which is equal to $N\phi_B$ for a closely wound coil and

in such a case,

$$N\phi_B \propto I$$

The constant of proportionality in this relation is called 'Inductance'.

➤ Inductance depends only on the geometry of the coil and the intrinsic material properties.

➤ inductance is a scalar quantity.

it has dimensions [$ML^2T^{-2}A^{-2}$]

SI unit is Henry (H)

○ Mutual Inductance-

✓ An electric current can be induced in a coil by flux change produced by another coil in its vicinity, this phenomenon is known as 'mutual induction'

✓ Inner solenoid S_1 and outer solenoid S_2 ,
When a current I_2 is set up through S_2 , it in turn sets up a magnetic flux through S_1 that is ϕ_1 ,

$$N_1\phi_1 = M_{12}I_2$$

M_{12} is mutual inductance of solenoid S_1 with respect to solenoid S_2 .

It is also referred as '*coefficient of mutual induction*'.

✓ For simple coaxial solenoids it is possible to calculate M_{12} .

$$N_1\phi_1 = (n_1l) (\pi r_1^2) (\mu_0 n_2 I_2)$$

$$M_{12} = (\mu_0 n_1 n_2 \pi r_1^2 l)$$

✓ If a medium is present then,

$$M = \mu_r \mu_0 n_1 n_2 \pi r_1^2 l$$

✓ Mutual inductance of a pair of coils, solenoids, etc., depends on their separation as well as their relative orientation.

✓ Varying current in a coil can induce emf in a neighbouring coil which is given by,

$$\varepsilon_1 = -M dI_2/dt$$

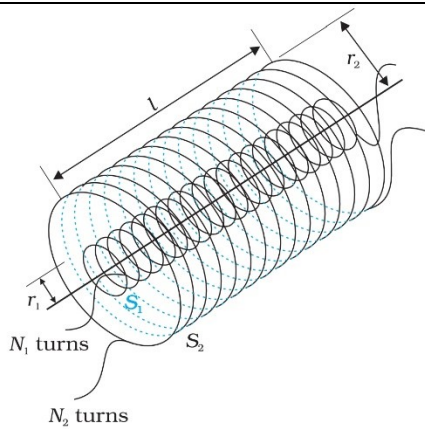


FIGURE 6.15 Two long co-axial solenoids of same length l .

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○ Self Inductance-

- ✓ **Emf is induced in a single isolated coil due to change of flux through the coil by means of varying the current through the same coil. This phenomenon is known as ‘self induction’.**

- ✓ **Flux linkage,**

$$N\phi_B \propto I$$

$$N\phi_B = LI$$

Here, constant of proportionality L is called ‘self inductance’

- ✓ **Induced emf,**

$$\varepsilon = -L \frac{dI}{dt}$$

self induced emf always opposes any change of current in the coil.

- ✓ **Self inductance of solenoid,**

$$L = \mu_0 n^2 A l$$

If a medium is present then,

$$L = \mu_r \mu_0 n^2 A l$$

- ✓ **L depends on its geometry and on the permeability of the medium.**

- ✓ **The self induced emf is also called *back emf* as it opposes any change in current in a circuit.**

- ✓ **The energy stored in solenoid,**

$$E = \frac{1}{2} LI^2$$

❖ AC Generator-

- In an ac generator , mechanical energy is converted into electrical energy by virtue of electromagnetic induction.
- If coil of N turns and area A is rotated at ν revolutions per second in a uniform magnetic field B, then the motional emf produced is,

$$\varepsilon = NBA\omega \sin(\omega t)$$

where, $\omega = 2\pi\nu$

(here , we have assumed that at $t=0$, the coil is perpendicular to the field .



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SUMMARY

1. The magnetic flux through a surface of area **A** placed in a uniform magnetic field **B** is defined as,

$$\Phi_B = \mathbf{B} \cdot \mathbf{A} = BA \cos \theta$$

where θ is the angle between **B** and **A**.

2. Faraday's laws of induction imply that the emf induced in a coil of N turns is directly related to the rate of change of flux through it,

$$\varepsilon = -N \frac{d\Phi_B}{dt}$$

Here Φ_B is the flux linked with one turn of the coil. If the circuit is closed, a current $I = \varepsilon/R$ is set up in it, where R is the resistance of the circuit.

3. Lenz's law states that the polarity of the induced emf is such that it tends to produce a current which opposes the change in magnetic flux that produces it. The negative sign in the expression for Faraday's law indicates this fact.
4. When a metal rod of length l is placed normal to a uniform magnetic field B and moved with a velocity v perpendicular to the field, the induced emf (called motional emf) across its ends is

$$\varepsilon = Blv$$

5. Inductance is the ratio of the flux-linkage to current. It is equal to $N\Phi/I$.

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6. A changing current in a coil (coil 2) can induce an emf in a nearby coil (coil 1). This relation is given by,

$$\varepsilon_1 = -M_{12} \frac{dI_2}{dt}$$

The quantity M_{12} is called mutual inductance of coil 1 with respect to coil 2. One can similarly define M_{21} . There exists a general equality,

$$M_{12} = M_{21}$$

7. When a current in a coil changes, it induces a back emf in the same coil. The self-induced emf is given by,

$$\varepsilon = -L \frac{dI}{dt}$$

L is the self-inductance of the coil. It is a measure of the inertia of the coil against the change of current through it.

8. The self-inductance of a long solenoid, the core of which consists of a magnetic material of relative permeability μ_r , is given by

$$L = \mu_r \mu_0 n^2 A l$$

where A is the area of cross-section of the solenoid, l its length and n the number of turns per unit length.

9. In an ac generator, mechanical energy is converted to electrical energy by virtue of electromagnetic induction. If coil of N turn and area A is rotated at ν revolutions per second in a uniform magnetic field B , then the motional emf produced is

$$\varepsilon = NBA (2\pi\nu) \sin (2\pi\nu t)$$

where we have assumed that at time $t = 0$ s, the coil is perpendicular to the field.

POINTS TO PONDER

1. Electricity and magnetism are intimately related. In the early part of the nineteenth century, the experiments of Oersted, Ampere and others established that moving charges (currents) produce a magnetic field. Somewhat later, around 1830, the experiments of Faraday and Henry demonstrated that a moving magnet can induce electric current.
2. In a closed circuit, electric currents are induced so as to oppose the changing magnetic flux. It is as per the law of conservation of energy. However, in case of an open circuit, an emf is induced across its ends. How is it related to the flux change?
3. The motional emf discussed in Section 6.5 can be argued independently from Faraday's law using the Lorentz force on moving charges. However, even if the charges are stationary [and the $q(\mathbf{v} \times \mathbf{B})$ term of the Lorentz force is not operative], an emf is nevertheless induced in the presence of a time-varying magnetic field. Thus, moving charges in static field and static charges in a time-varying field seem to be symmetric situation for Faraday's law. This gives a tantalising hint on the relevance of the principle of relativity for Faraday's law.

